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Integrating Personalized and Accessible Itineraries in MaaS Ecosystems Through Microservices

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Abstract Mobility is a crucial sector for the livability of urban spaces, both in terms of accessibility for people with disabilities, and in terms of enjoyability by people with different interests. The deep transformation mobility is undergoing, heading towards commoditization of the full spectrum of transportation services, can lead to efficient solutions based on the same principle for all these needs. This paper shows how the approach based on the flexible orchestration of microservices allows to build applications that are, at the same time, more easily suited to the specific needs of different user categories, and more seamlessly integrated in the Mobility as a Service approach to smart mobility.

Keywords Mobility-as-a-Service · Microservices · Users with special needs

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1 Introduction

The advent of smart mobility has enabled the creation and development of applications for mobility support addressed to different categories of people, including those with disabilities [1, 2]. Yet, the majority of these development paradigms provides vertical applications for accessible mobility, separated from the wealth of services available for average-capability travelers, which in turn are often competing on quality of information and interfaces rather than customization. The integration of collected data, the possibility to share common sub-services, and the ability to interact with the same standards to provide highly configurable results is rather difficult under the more traditional business models and software development approaches.

On the other hand, the rapidly growing diffusion of ICT solutions in every field of mobility is fostering the creation of novel approaches, such as the so-called *Mobility-as-a-Service* (MaaS) paradigm [3]. The enabling factor for these approaches to succeed is a strong drive towards standardization of data formats and access methods. According to the most modern model, every access to a piece of data, as well as every execution of a simple computation is offered as a service [4]. The collection of domain-specific services, completed by a set of commonly used infrastructural services, composes a service-oriented architecture, on which it is comparatively simple and fast to develop any application by orchestrating the needed services [5].

The purpose of this paper is to show the potential of a platform based on microservices for the development of scalable and integrated mobility services. Two use cases demonstrate how both accessibility features and generic user preferences can be taken into account with the same system architecture. The paper is structured as follows. Section 2 provides an overview of the relevant literature

for the technologies and models exploited in the proposed platform. Section 3 illustrates the platform architecture, focusing in particular on the advantages that this paradigm introduces on the basis of a real, currently developing infrastructure, which is called *Smart Mobility for All* (SMAll). Then, Section 4 details our use cases developed in the urban region of the city of Bologna. Finally, Section 5 concludes the paper with some final remarks and directions for future work.

2 Background and related work

In this section, we briefly describe main literature where crowdsourcing and crowdsensing are exploited in smart city contexts, with the aim of improving citizens quality of life within urban environment, as well as the modern approaches that support smart mobility, namely Mobility as a Service and microservice-based architectures.

2.1 Crowdsensing and crowdsourcing

Crowdsensing can play a key role in smart cities, acting as a technological enabler [6]. It gets benefits from the wide diffusion of mobile devices (equipped with different kind of sensors, such as gyroscope, GPS, compass, etc.) and from a massive users' participation [5, 7]. In particular, crowdsensing can be exploited by involving sensors which are naturally moving (i.e., sensors are mounted on mobile devices, carried by citizens across the urban environment) [8]. Using moving sensors in crowdsourcing systems is called *mobile crowdsensing* (MCS) [9]. In particular, MCS involves people who are moving and collecting data from different places and paths, by carrying sensors that are integrated to mobile devices and by providing information about the surroundings manually too.

With particular regards to smart mobility, crowdsensing and crowdsourcing can support urban transportation optimization [10]. In fact, traffic can be unpredictable, because it could be influenced by rush hours, accidents, weather conditions. All theses issues can affect vehicles speed while traveling in the city. In order to be effective, services based on crowdsensing and crowdsourcing have to reach a critical mass of collected data [11]. Hence, users should be felt motivated and involved in the activities of gathering data [12]. Engaging more contributors in participating corresponds to an increment of the quality and the quantity of collected data [13]. Among all the crowdsensing and crowdsourcing systems and services designed and implemented within the smart city paradigm, some are devoted to support citizens participation and collaboration in improving the quality of life in urban environments [14]. And a part of them gathers data about urban accessibility [15], with the specific aim of improving the quality of life and the independence of people with disabilities [16].

In this special context, sensing apps devoted to monitor human activities can be customized so as to detect accessibility barriers (e.g., stairs) and facilities (e.g., zebra crossing) in the urban environment [17].

In [18], the authors exploit data obtained by a smartphone accelerometer with the aim of identifying the position where a pedestrian stops and crosses a street ruled by a traffic light. In similar contexts, cooperative sensing can be used to detect movement of groups of people [19]. Some methodologies useful in developing a large accessibility map with personal sensing through smartphones have been proposed in [20]. In particular, this work is based on the idea of exploiting devices held by wheelchair citizens and then to apply machine learning technologies (i.e., supervised learning techniques) to estimate types of ground surfaces.

Among sensing technologies, we deem useful to describe acoustic recognition in more detail. First described in [21], it is a technique for recognizing acoustic properties through a combinations of several temporally-and spectrally-based analysis of different acoustic bands. This properties are used to create a footprint of a specific environment.

As described in [22] this properties can be used as a method for classifying environment types based on acoustic signals. Other work as [23] used this classification to collect an audio database, which contains recordings from a variety of different auditory contexts. It is also proven in works as [24] and [25] that these sets can be used to recognize new environmental sounds for the understanding of a scene or context.

There are several possible usages of this techniques in the actual network of sensing services: new generation of sensors, robots, military contexts, etc. can benefit of this feature. As described in [23] practical applications can include intelligent wearable devices and hearing aids that sense the environment of their users at a given time. The information about the environment enables the device to provide better service to users' needs.

In the context of smart and accessible mobility, acoustic recognition can complement typical location information from wifi hotspot, gps and so on, to achieve not only a positioning improvement when traditional means are imprecise (typically, indoor), but also context awareness [26] that can be used in many situations, for example, a GSM phone may know not to ring when the user is at a lecture or in a meeting, and in contrast, it may ring louder when the user is in a noisy place.

2.2 Mobility as a service

Another concept at the basis of our work is Mobility as a Service (MaaS) [3]. One of the main advantages of a

MaaS provider is that it shall offer a unique and seamless interface to users, aggregating heterogeneous transport options offered by different mobility providers (e.g., different agencies providing transportation by taxi, bus, train, airplane, car-sharing, etc., including the public transportation providers), handling the whole experience of traveling, from providing information, to travel planning, and payments [27].

Very briefly, the principle of MaaS is that as long as every detail of the demand and supply for transportation services is known in real-time, there is no need for passengers to commit on specific means. Instead, they will enjoy a broad spectrum of alternatives from which to choose, taking into account the needs of the moment. For example, one could specify a very strict set of constraints in terms of comfort and timing, likely to result in a choice of premium means, while another could simply express the need for reaching a destination at the best price, getting a virtual ticket, and receiving real-time instructions about which means to use to complete the trip. Many business models are possible [28], and many kinds of public/private partnerships can be imagined. In its simplest form, a MaaS operator is simply a smart broker pooling transport options. Service customization and capacity trading can be pushed to very sophisticated levels, and public administrations can take up new roles, pursuing their goals (e.g., development of some areas, congestion and pollution control, etc.) by means of either direct intervention on the technological platform or indirect regulation of the MaaS market.

Typically, scenarios with few users involved, and that require specific technologies, as it happens in the case of people with disabilities, have always been profitable only through a heavy vertical investment [29]. Thanks to the MaaS principle, where single services can be shared and public administrations can direct investments on the development of specific needs, even this kind of use case can be seen as an opportunity for MaaS operators.

2.3 Microservices

The microservice architecture is a development style inspired by service-oriented computing that has recently started gaining popularity [30]. A microservice architecture can be defined as: "a distributed application where all its modules are microservices." [31]. The reasons why this architecture has been proposed is to try to overcome the limits of the so-called monolithic architecture [32, 33], which, as the software grows in size, become hard to maintain and to deploy at scale. The microservice architecture has been proposed with a diametrically opposite approach. Rather than having a single block composed of several interdependent modules, the microservice philosophy extracts each module making it an independent and autonomous microservice following the statement "everything can be a microservice" [34].

This kind of distributed compositionality becomes his greatest strength. As described in [31], even the internal components of software are autonomous services, independent components conceptually deployed alone with dedicated technologies, both software and hardware. This means that this architectural style does not foster or forbid any particular programming, but it provides only a partition structure of the component the developer has to follow.

In addition, we have to consider the fact that this paradigm is a reasonable consequence of a world that is more and more oriented to the development on cloud, and therefore the need of connection and interoperability between the different platforms is increasing.

Since all the components of a microservice architecture are microservices, its distinguishing behavior derives from the composition and coordination of its components via messages. Thus, the core of a microservice platform will be the orchestration phase, namely the composition of microservices, tools, and processes invoked, and the connection and automation of workflows that create the final service [35].

However, the microservice paradigm introduces also new challenges and enlarges known threats. The distributed programming nature is at the same time its main advantage (as shown earlier) and disadvantage. As described in [36], preventing programming errors in this context is harder, and the programmer has to face and work with different technologies that typically have different means of specifying contracts for the composition of services.

Other issues introduced by the microservice architecture are the ones related to the security aspects. These issues are certainly not new, as they apply to SOA and in general to distributed computing, but they become even more challenging in the context of microservices. Description of this kind of issues can be found in [37–40].

3 A microservice-based platform

To effectively support the creation of MaaS providers, we envision the creation of an ecosystem of reusable components within a Service-Oriented Architecture, where standardized services efficiently and flexibly combine heterogeneous data sources, such as available transport options, real-time data regarding vehicles and infrastructures, pricing, etc., to provide customized travel planning, information and ticketing to final users, as well as monitoring and planning tools to policy-makers.

This infrastructure aims to create services to track timing, position, and availability of trains, buses, subways, shared bikes, shared cars, and to enable crowdsourcing of users'

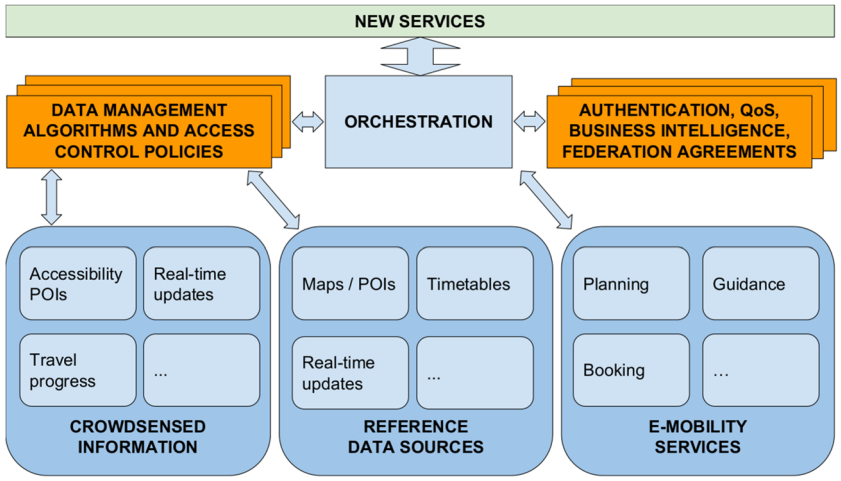

Fig. 1 A microservice platform to enable the creation of mobility services

report, crowdsensing through devices [18, 41], etc. in an overall effort of opening data and standardizing the interfaces to access them. Both operators and users can benefit from this plethora of mobility services.

A real example of this kind of infrastructure that support the creation of such a liquid market of transportation is the one that we are currently developing as a marketplace for mobility services, within the EIT Digital[1] project called *Smart Mobility for All* (SMAll) [42]. In our vision, SMAll is the enabling technology that facilitates the publication, retrieval, and orchestration of functionalities for mobility, provided by different mobility operators, solving the challenges of the MaaS market, from developing user-contributed, crowdsourced applications, to launching a MaaS operator, to planning effective and sustainable transport policies for smart cities [15]. SMAll builds on the concept of Federated Cloud Computing [43, 44] and maintains an open approach with respect to the possible members of the federation: the platform is open not only to MaaS operators but also to traditional transport agencies and even players that provide information like weather forecasts or crowd-sourced data [45] for mobility. In these regards, and as envisioned in [46], SMAll can play an important role in supporting the planning of sustainable and participated transport policies for smart cities.

The choice on microservices comes from the consideration that operators already have a collection of legacy components that address some specific issues (e.g., the Bus GPS Positioning and the Timetables services) and that other operators are willing to pay to access them. In practice, operators would like to include in their own services some specific external component (e.g., a bus tracking service) rather than use (and pay for) a bundle of unneeded functionalities (e.g., a real-time planner that includes the mentioned bus tracking capability). Microservice architectures achieve such degree of granularity.

As expected, microservices come with some trade-off: loosely-coupled microservices communicate via message passing, which requires proper routing and may suffer latencies and failures; many requests can overload a service, hence it should prevent outages by limiting them and/or scale accordingly; services are heterogeneous but their data-formats and Application Programming Interfaces (APIs) should be homogeneous to foster compositionality.

For these reasons, SMAll strives for standardization of data-formats and APIs. Moreover, it provides infrastructural components to cope with most of the aforementioned issues: orchestration abstractions to streamline the composition of available services (via the Jolie programming language [47, 48]); data-format conversion functionalities; service registries and dispatchers to store the definition and address of all the services deployed on the platform and to route requests to them; business intelligence outlets for auditing and performance indicators on the usage of services.

In Fig. 1 is represented the SMAll model of such a microservice platform [5].

4 Use cases

In this section we describe two specific use cases which have been developed by means of the SMAll platform. The goal of such use cases is to show how the microservice architecture can be exploited in the context of smart mobility, meeting different kinds of users' needs.

We present two scenarios involving the entire stack of sensors and technologies of a personal smartphone for the support of (i) a tourist activity, and (ii) a blind user who requests an accessible path, detailing the orchestration of a pool of microservices, each one in charge of handling a specific mobility need.

[1] http://www.eitdigital.eu/

In both the scenarios kept into account, the user asks for a specific multimodal path (including bus routes) starting from the School of Engineering and Architecture to reach the School of Art, Humanities and Cultural Heritage of the University of Bologna. The most commonly used geospatial mapping platforms (such as Google Maps) usually propose the shortest path (the black one shown in Figs. 2 and 4). Such a path (expected to take 18 minutes) is structured in three parts, as follows:

1. a walking part to reach the bus stop; this part is supposed to take 4 minutes (for a 300 meters distance);
2. a part of a bus route (between the bus stop icons); this part is supposed to take 8 minutes (with five inbetween stops);
3. another walking part from the arrival bus stop to the final destination; this part is supposed to take 6 minutes (for a 500 meters distance).

The result of the microservices orchestration is a workflow where, a crowdsourcing and crowdsensing microservice helps the planning, the traveling and the reporting phases in providing a personal trip in a urban context, taking into account the user' needs and preferences.

4.0.1 Microservices involved

The main features needed to build an application for such a kind of use cases can be grouped in three different sets of microservices: crowdsourcing, crowdsensing, and route planning. Indeed, an important role is also played by the service involved in the user' preferences settings and management. The crowdsourcing microservice is based on a set of push notifications (to check the trustworthiness of collected data) and a voluntary reporting system (to collect data) that will allow the user directly interact to and inject several information into the platform. Thanks to this microservice, users can populate the SMAll datasets, gathering data related to:

– POIs (Point of interests) such as monuments, public services stops, parks, historic sites and so on;
– aPOIs (accessibility Point of interests) such as barriers (stairs, gaps, obstructions, etc.) and facilities (curb cuts, traffic lights, zebra crossings, etc.);
– information about street and route availability, public services timetables and time schedules.

The crowdsensing service works automatically acquiring different kinds of information from different sources; this includes:

– realtime gps information and realtime access point information;
– accelerometer and gyroscopic information to detect aPOI such as stairs and gaps.
– acoustic frequencies for acoustic recognition.

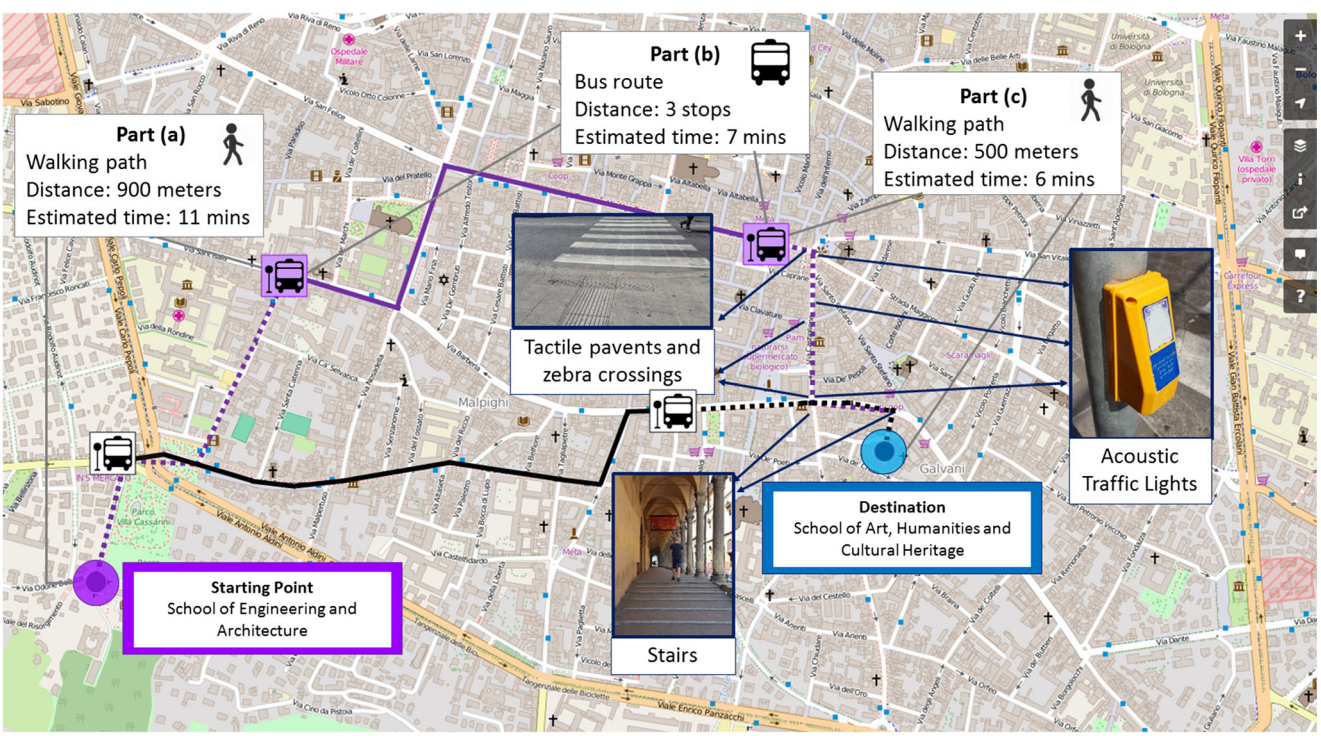

Fig. 2 The personalized path between the starting point and the destination

The acoustic recognition feature will be used to improve the accuracy of the user location. This will be done mainly for two specific situations:

- specific recognition in indoor environment;
- improvement of the position in tight spaces like alleyways between buildings etc.;
- increment of the orientation and confidence while walking in outdoor environment leveraging on the recognition of well-known sounds (as the noise of buses in a bus station, the kids clamour close to a school, and so on).

This is an extremely important help because it significantly improve the Position Tracking in areas where precision is usually less than 10 meters. It is known in the literature fact that at present the position of the GPS does not go below 10 meters due to physical limitations, environmental and deliberately introduced by the American military command which held greater effectiveness for military purposes only [49].

The practical cases are manifold. Take for example a blind user to be guided inside a museum or facility in which the GPS signal or hotspot is weak or obstructed. In this case the acoustic recognition helps the environment tracking. Similarly, a travel planner can greatly improve tracking of the position under special conditions such as narrow alleys between buildings as "high", that is where the degree of reliability of the GPS is not enough to ensure full support to user in particular context. Moreover, some well-known sounds (as the noise of buses in a bus station, the kids clamour close to a school, and so on) can help user in orienting (and in acquiring confidence about) the walked path. This feature can be very useful for users with different kinds of disabilities, elderly people and kids.

As router planning, we exploited the open-source OpenTripPlanner engine [50] to compute routes between the points of interest specified by the user. OTP allows users to compute multimodal paths, combining multiple modes of transportation. We customized the OpenTripPlanner routing engine, including the possibility to compute the path keeping into account the user's needs and preferences in terms of POI, aPOI, and so on.

The SMAll platform is able to orchestrate all the presented microservices, which relying on several data sets and on different data typologies, in a uniform and transparent fashion.

4.1 First scenario: a blind user

In the first scenario we present a use case illustrating a blind user who requests a personalized path along the city of Bologna (Italy), by using his own smartphone. In particular, let us consider a male undergraduate student equipped with a white cane who uses to move across Bologna from home to reach the University campus and among offices and classrooms of the University, which are spread all over the city. He has set up his profile with preferences related to the crossing facilities (he LIKEs zebra crossings, traffic lights, and audible traffic lights, since it is much easier for him crossing streets by exploiting such urban elements), surfaces (he LIKEs tactile pavings, which help him in walking on the street in an independent and safe way), bus and bus stops characteristics (he LIKEs tactile information and acoustic cues and announcements), and other urban elements (such as path-ways, obstructions and so on). He LIKEs steps and stairs because they represent a landmark, helping him in orientating while he is wandering the city.

Taking into account the preferences already discussed, the shortest path depicted in Fig. 2 (the black one) presents some issues our blind citizen has to face: (i) the absence of tactile pavings and of acoustic traffic lights; (ii) the absence of tactile information and of acoustic cues at the bus stops; (iii) the absence of acoustic announcements and of tactile information on the bus of that line; (iv) the information about bus arrival time is derived from a time table, instead of referring to the real bus position and availability.

When our user asks for a path from the starting point to the destination, then our system computes a personalized route taking into account the user's profile (i.e., avoiding those barriers which affect him and including as much as possible the LIKEd facilities).

Our system computes a personalized path, by taking into account real data about bus availability and the user's profile, in terms of barriers to avoid and LIKEd facilities to include as much as possible. In particular, our system matches the user's preferences with the information about the aPOIs (accessibility Points of Interest [1]) all over the city and real time open data about the public means of transport. A personalized path that meets these issues, computed by our system is shown in Fig. 2.

Such a personalized path is longer than the original one, and it is expected to take 24 minutes. According to a survey [51], citizens would face a longer path in urban environments if it meets their preferences. In particular, 88 % of them were ready to face a path up to 30 % longer to reach their destination if such a route was tailored to their preferences and needs, while 12 % of the users would face a personalized path more than 30% longer.

The personalized path of our use case is structured in three parts, which involve a different bus line in its second part. In particular, such a path lets the user exploits urban facilities he declared as "LIKE" in his profile (e.g., zebra crossings, tactile pavings, acoustic traffic lights, bus stops equipped with tactile and acoustic cues), which improve his independence in moving across the city. During the third part of his personalized route, the user meets two stairs.

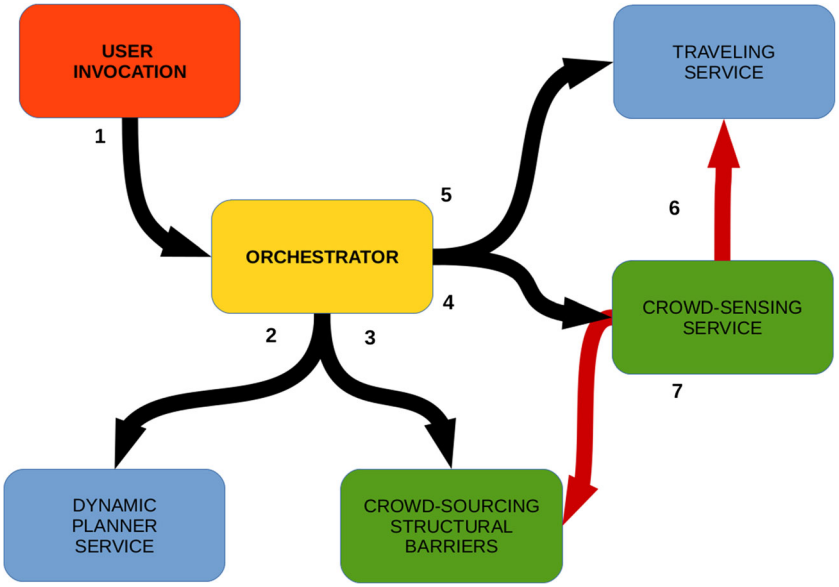

Fig. 3 The use case workflow represented as a microservice architecture in SMAll in the first scenario

Incidentally, his smartphone senses them, thanks to our system, making the user also a provider of sensed data related to urban barriers.

A representation of the workflow of this use case in a microservice architecture is shown in Fig. 3. The user, in our case the blind user, invokes the routing service (1). The orchestrator intercept this call and it is the one in charge to create the correspondent workflow based on the invocation parameters. The invocation parameters are represented as the profile LIKES, and the orchestrator invokes the routing algorithm of the dynamic planner service (2) and the crowdsourcing service that maps the structural barriers (3) based on the defined preferences. Once the trip has been proposed and chosen, the orchestrator calls at the same time the crowdsensing service (4) and the traveling one (5). The traveling service manages the assistance phase, guiding the blind user and exposing the information with in an appropriate manner, e.g., voice synthesizer. The crowdsensing

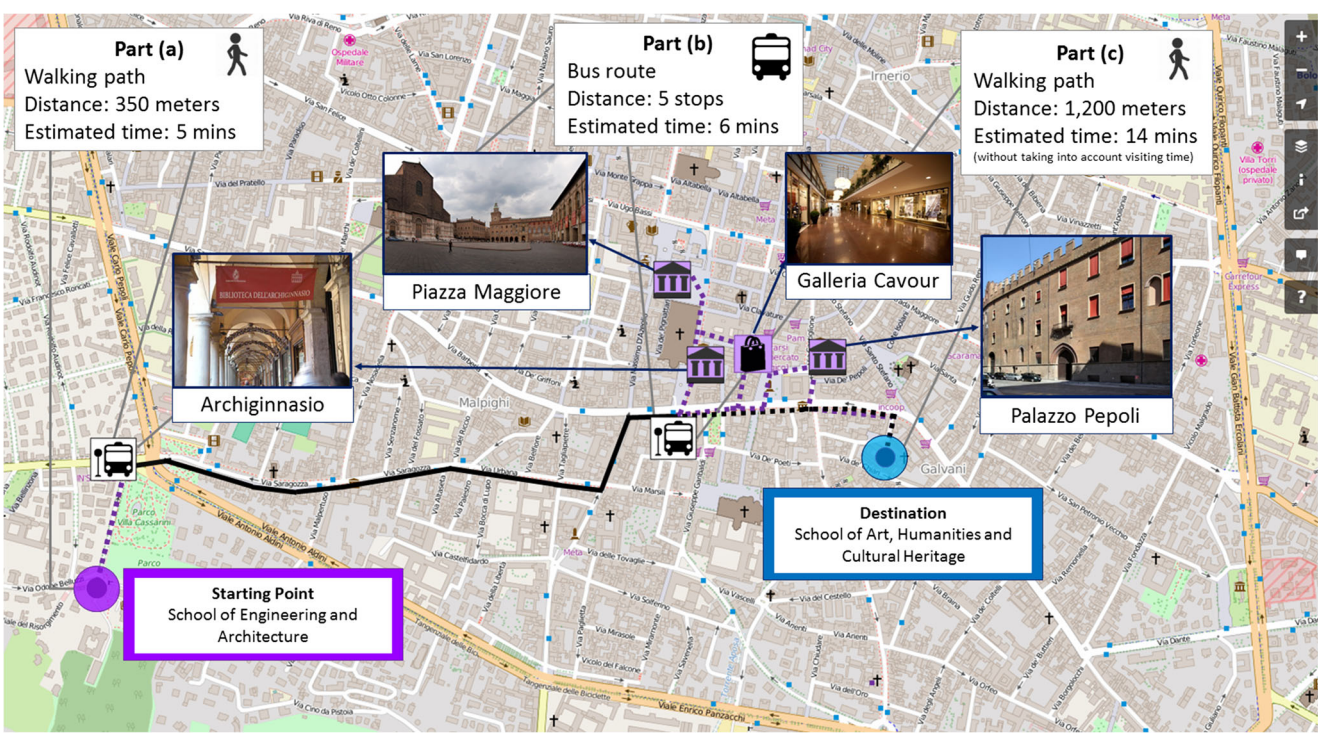

Fig. 4 The shortest path between the starting point and the destination (in black) and the customized path in the first use case

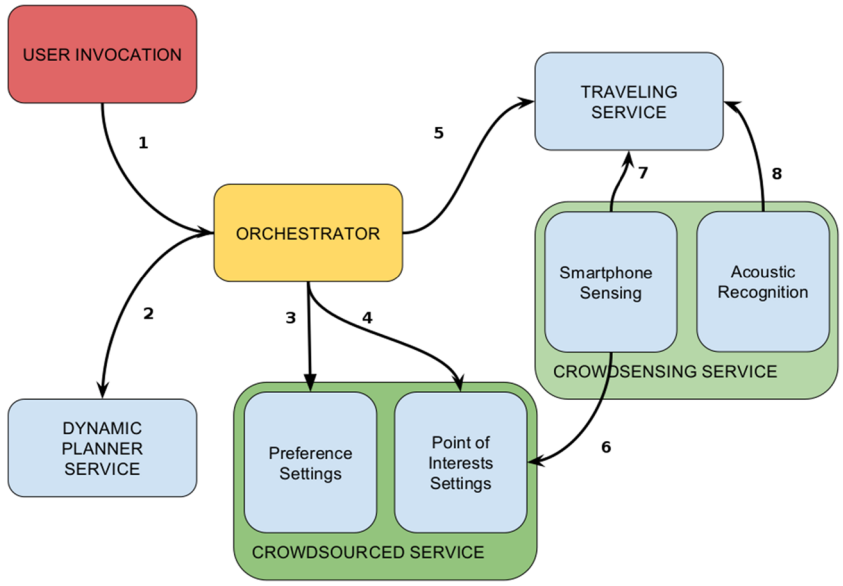

Fig. 5 The use case workflow represented as a microservice architecture in SMAll in the second scenario

service otherwise has a twofold goal. Based on the hardware device used, it will collect (possibly) new information about the path in place at that time, e.g., such as the presence of stairs, obstacles or other temporary interruptions. This information will be used to immediately interact with the user notifying him warnings or advisories and, at the same time, this information will feed the crowdsourcing service enriching its database for future requests.

4.2 Second scenario: a tourist user

The main idea of this scenario is to show how a general tourist, an adult woman, thanks to our platform, can set up a personalized environment in order to find the best ”tourist” path according to her preferences.

The first step for the tourist user is to set up the profile preferences in term of LIKE, DISLIKE, AVOID and NEUTRAL. Once she has set up her profile with preferences related to POIs, the system compute the best paths that meets the user expectations. In particular, this tourist is interested in (LIKE) historical building and monuments, museums related to the history of the city and exclusive shopping centers. She does not care (NEUTRAL) about parks and green areas and she does not like (DISLIKE) noise pollution areas. Each preference is related to different set of data that are collected thanks to the crowdsourcing and crowdsensing microservices.

The shortest path, calculated without keeping into account the user preferences, does not include any interesting attractions, and is the black path shown in Fig. 4. Instead, the one provided by our platform, refers to different POIs, located a short walk away from the user' destination and is shown as the purple path, the dashed lined, in Fig. 4 as a pursuance of the black path one. In this area, the tourist, exploiting the crowdsourcing service can collect updated information related to historical building and monuments and share comments and notes. Moreover, when inside to a tourist building, the smartphone equipped with our application, can activate the acoustic recognition feature for recording sounds or playing vocal comments and indications related to the specific location.

Figure 5 represents the workflow of this use case in a microservice architecture. The user, in this case the tourist user, invokes the routing service (1). The orchestrator intercept this call and it is the one in charge to create the correspondent workflow based on the invocation parameters, represented as profile LIKES/DISLIKE/NEUTRAL issues. The orchestrator invokes the routing algorithm of the dynamic planner service (2) and the crowdsourcing service that maps the POIs (3) based on the user' preferences. Once the trip has been proposed and choose, the orchestrator calls at the same time the crowdsourcing service (4) and the traveling service (5). The crowdsourcing service has a twofold goal. First of all, it allows the user to collect information related to the visited places. Second, it maps information (gathered by other users) about tourist places and historical building, as requested by the user. The traveling service manages the navigation phase (indoor and outdoor) interacting with the crowdsensing microservice and the acoustic recognition feature.

5 Conclusion

Smart mobility is a key element to support citizens in their daily activities and to offer them a livable smart city. Information about urban transportation (including taxis, bus,

train, car-sharing, etc.), urban barriers and facilities, pedestrian and multimodal paths would be of great benefit in this context, as well as all the information about the whole experience of traveling and wandering the city, including travel planning and payments. In order to provide a platform to manage such services and features, we designed and prototyped an infrastructure as a marketplace for mobility services, called Smart Mobility for All (SMAll). A prototype of such infrastructure has been developed and its microservice architecture has been described in the paper. Two different use cases, involving a blind user and a tourist moving across the urban environment, are detailed in the paper and they show the feasibility and the potentialities of our approach. Furthermore, the open nature of a microservice-based architecture allows the illustrated services to be integrated into any MaaS platform.

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